

Agenda

- Netflix Prize Solution
- MF feature engineering

Netflix prize competition

→ Metric: Reduce RMSE

$$\hookrightarrow \sqrt{\frac{1}{m} \sum (r_{ui} - \hat{r}_{ui})^2}$$

$$L = \min_{\substack{p_u, q_i \\ u, i \in K}} \sum \left(r_{ui} - \underbrace{q_i^T p_u}_{\hat{r}_{ui}} \right)^2 + \lambda_1 \|q_i\|^2 + \lambda_2 \|p_u\|^2$$

✓ p_u = User vector in d-dimension

✓ q_i = Item vector in d-dimension

Linear Regression

$$L = \min_{w, b} \sum \left(y^{(i)} - (w^T x^{(i)} + b) \right)^2 +$$

Regularization
coefficient.

$$\lambda \|w\|^2$$

Regularization

$$w = w - \alpha \frac{\partial L}{\partial w}$$

$$L = \min_{q_i, p_u} \sum_{u,i \in K} (\hat{r}_{ui} - \underbrace{q_i^T p_u})^2 + \lambda (\|p_u\|^2 + \|q_i\|^2)$$

p_u = User vector in d-dimension

q_i = Item vector in d-dimension

r_{ui} = User - Item rating

λ = Regularization Coefficient.

$$W = W - \alpha \frac{\partial L}{\partial W}$$

$$\begin{cases} p_v = p_v - \alpha \frac{\partial L}{\partial p_v} \\ q_i = q_i - \alpha \frac{\partial L}{\partial q_i} \end{cases}$$

SGD

$$L = \min_{q_i, p_v} (n_{vi} - q_i^T p_v)^2 + \lambda (\|p_v\|^2 + \|q_i\|^2)$$

$\swarrow$ Regularization coefficient
 $\hookrightarrow p_v^T p_v$ $\downarrow q_i^T q_i$

$$e_{vi} = n_{vi} - q_i^T p_v$$

$$\begin{aligned} \checkmark \frac{\partial L}{\partial p_v} &= 2(n_{vi} - q_i^T p_v) \cdot -q_i + 2\lambda p_v \\ &= -2 e_{vi} q_i + 2\lambda p_v \end{aligned}$$

— (1)

$$\begin{aligned} \checkmark \frac{\partial L}{\partial q_i} &= 2(\underline{n_{vi} - q_i^T p_v}) \cdot -p_v + 2\lambda q_i \\ &= -2 e_{vi} p_v + 2\lambda q_i \end{aligned}$$

— (2)

User update

$$p_u = p_u - \gamma \frac{\partial L}{\partial p_u} \quad \text{learning rate}$$

Item update

$$= p_u + \gamma (2e_{ui}q_i - 2\lambda p_u)$$

$$q_i = q_i - \gamma \frac{\partial L}{\partial q_i}$$

$$= q_i + \gamma (2e_{ui}p_u - 2\lambda q_i)$$

- $q_i \leftarrow q_i + \gamma \cdot (e_{ui} \cdot p_u - \lambda \cdot q_i)$
- $p_u \leftarrow p_u + \gamma \cdot (e_{ui} \cdot q_i - \lambda \cdot p_u)$

Bias term

1. Average rating in ecosystem (μ) ✓
2. Average rating of each user (b_u) ✓
3. Average rating of each movie (b_i) ✓

$$\hat{r}_{ui} = q_i^T p_u \quad \checkmark$$

$$\hat{r}_{ui} = \underline{q_i^T p_u} + \mu + b_u + b_i \quad \leftarrow$$

$$3.7 = \underline{q_i^T p_u} + b_u$$
$$-0.3 + 4 \quad \checkmark$$

$$1.7 + \underline{2} \quad \checkmark$$

4 ✓

Onig

$$L = \min_{q_i p_v} \sum_{u,i \in K} \left(\tilde{r}_{ui} - \underbrace{q_i^T p_v} \right)^2 + \lambda \left(\|p_v\|^2 + \|q_i\|^2 \right)$$

Bias

$$L = \min_{q_i p_v} \sum_{u,i \in K} \left(\tilde{r}_{ui} - \left(\underbrace{q_i^T p_v}_{+ \mu} + b_u + b_i \right) \right)^2 + \lambda \left(\|p_v\|^2 + \|q_i\|^2 + \mu^2 + b_u^2 + b_i^2 \right)$$

Temporal dynamics

$$\textcircled{1} \quad \hat{r}_{vi} = \mu + \check{b_u(t)} + \check{b_i(t)} + \check{q_v^T i} \check{p_v(t)}$$
$$\min \sum (r_{vi} - \hat{r}_{vi})^2 + \lambda \text{~~~~~}$$

↓
User preference
change with
time

$$\textcircled{2} \quad \min \sum \check{\check{w_{ui}}} (r_{vi} - \mu - b_u - b_i - q_v^T i p_v)^2 + \lambda \text{~~~~~}$$

↓
Exponential
Smoothing

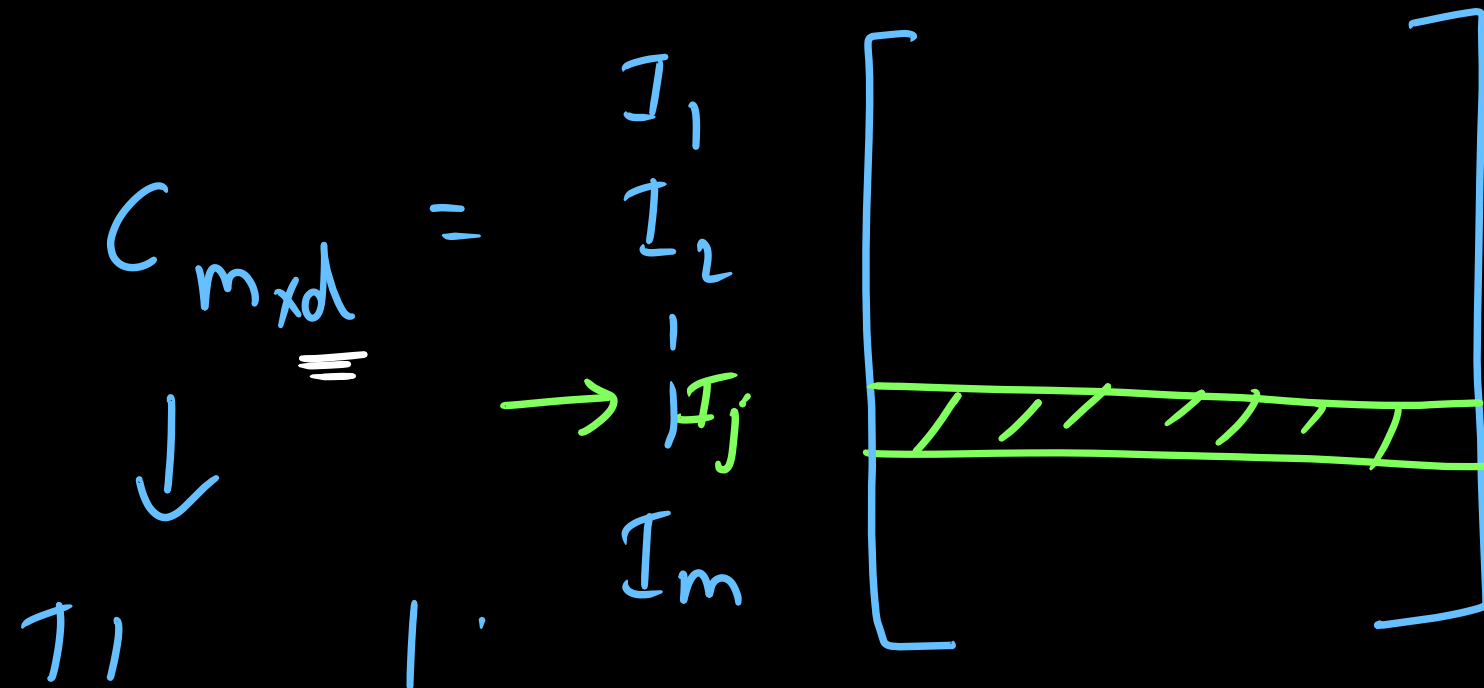
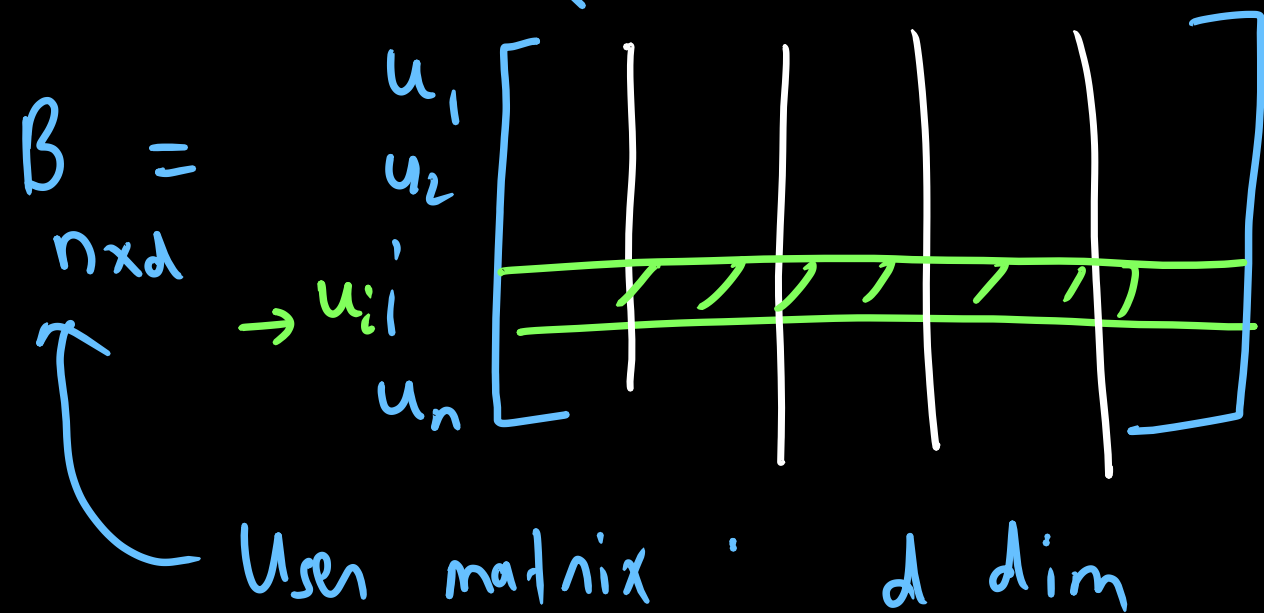
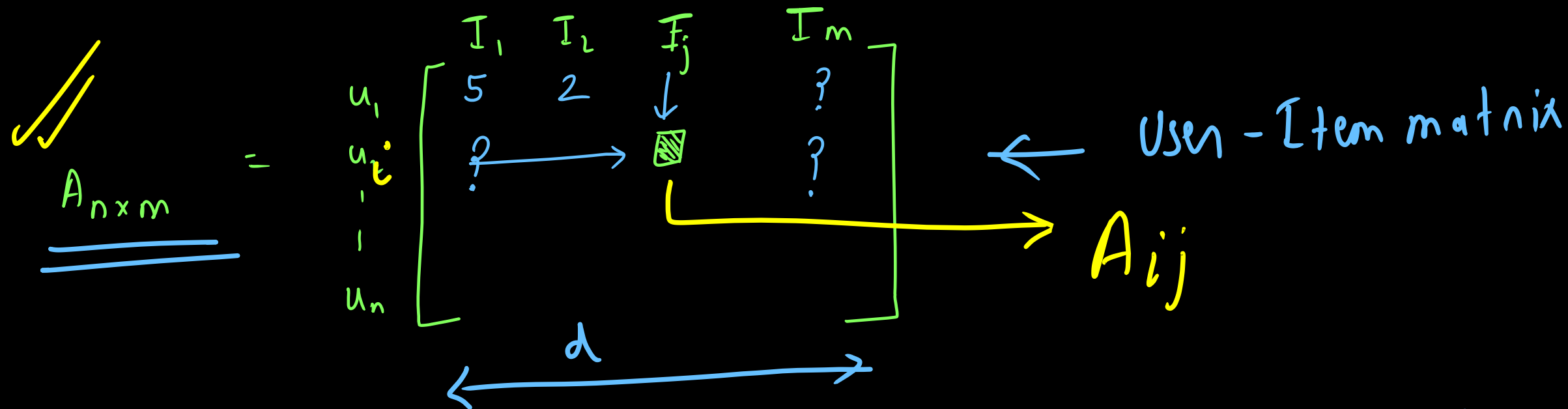
→ Break until 22:15 PM

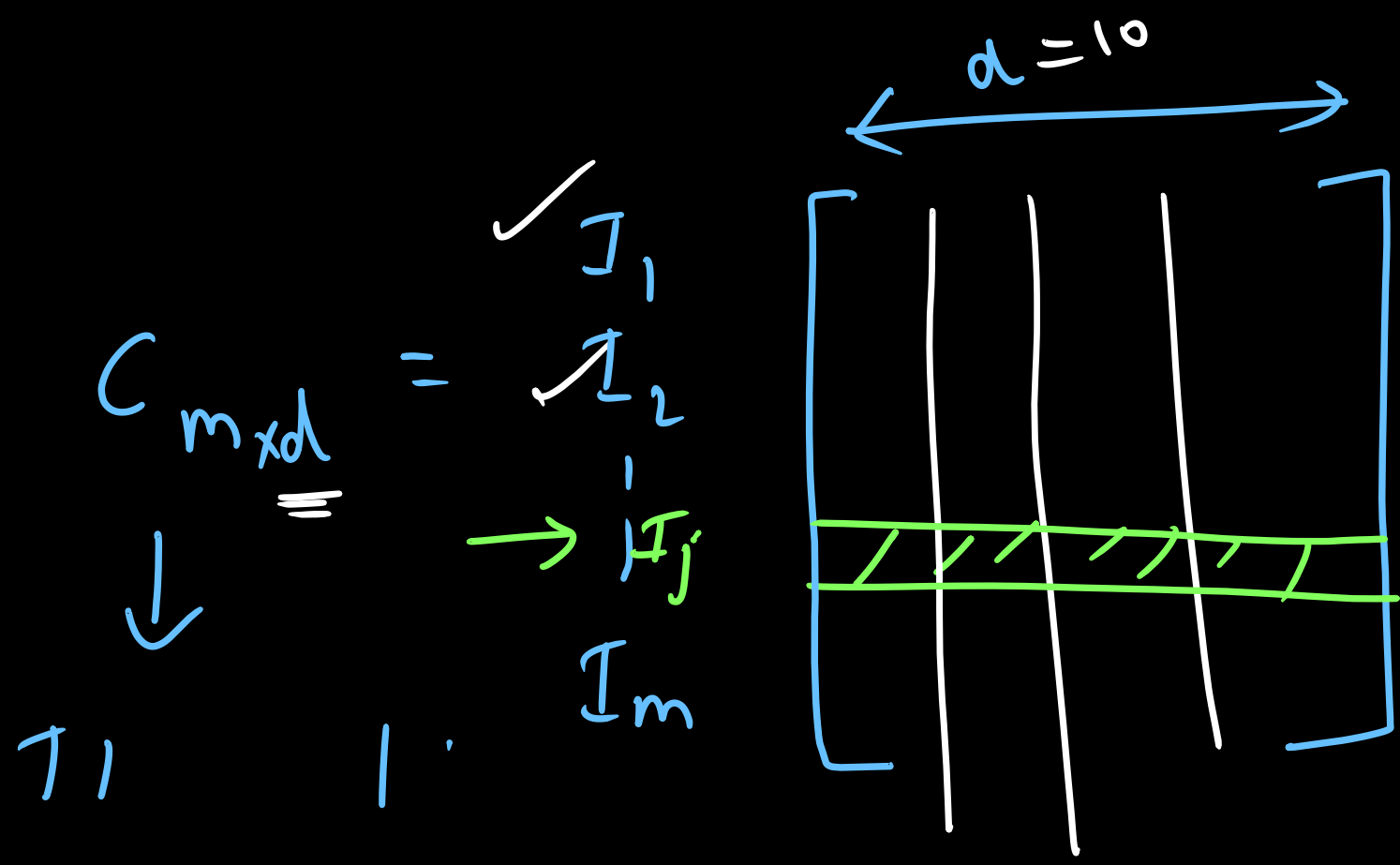
→ MF Feature Engineering

$$A_{n \times m} \approx B_{n \times d} C_{d \times m}^T$$

$$A_{ij} = B_i C_j^T$$

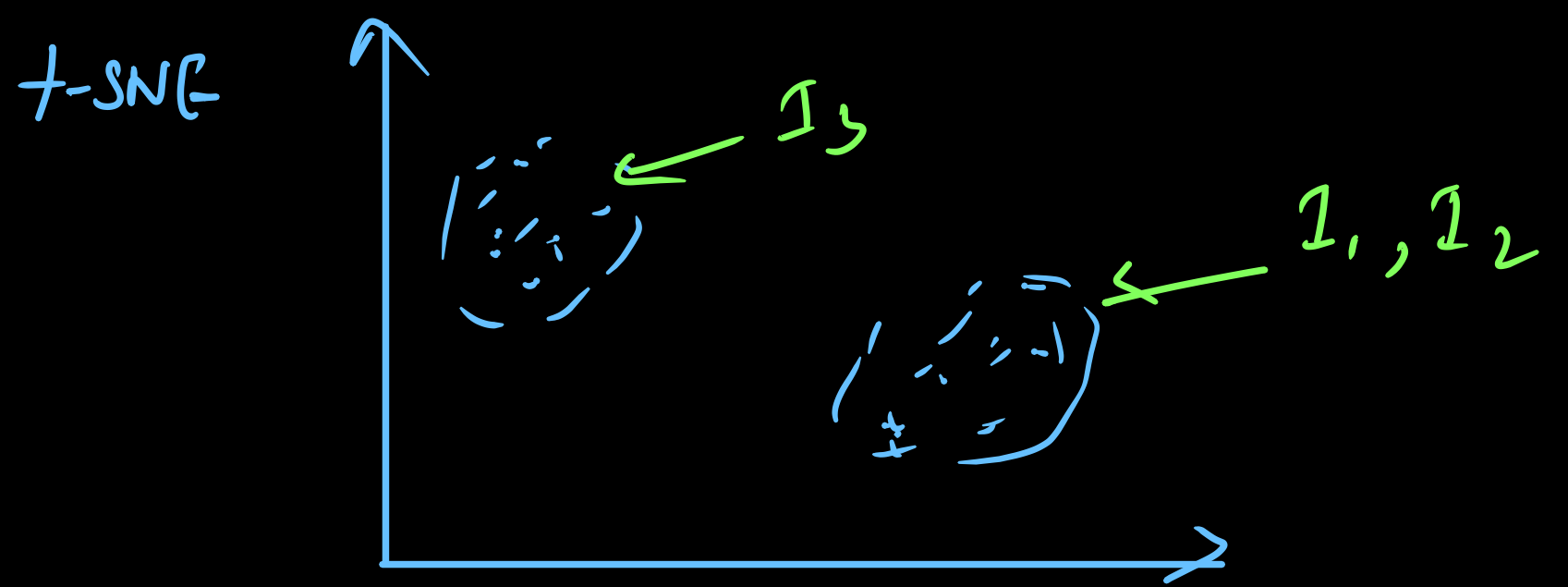
1×1 $1 \times d$ $d \times 1$



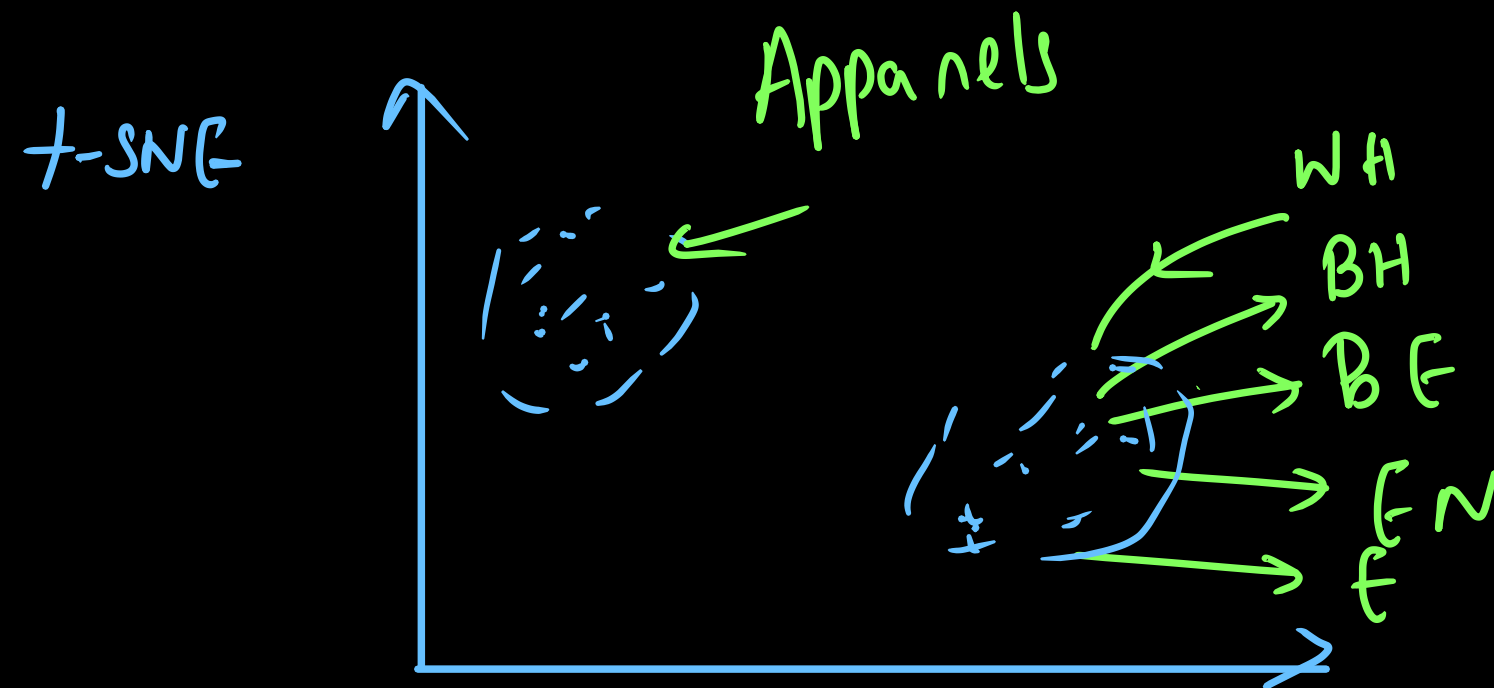


$\rightarrow$ Item vectors
 in d -dimension

$I_1, I_2 \quad \uparrow$
 $I_1, I_3 \quad \downarrow$



Query
bluetooth headphones $\rightarrow$ top-K similar items
 $\rightarrow$ Wired headphone
 $\rightarrow$ bluetooth earphone
 $\rightarrow$ Ear Neckband
 $\rightarrow$ Earpods (Airpod)



→ MF for text data

D1 : good boy good work

D2 : good girl

D3 : boy girl good

Vocabulary : { good, boy, work, girl }

↳ Unique word

→ MF for text data

→ D1: good boy good work ←

D2: good girl

D3: boy girl good

Vocabulary: {good, boy, work, girl} ✓✓

	good	boy	work	girl
D1	2	1	1	0
D2	1	0	0	1
D3	1	1	0	1

BOW
(Bag of Word)

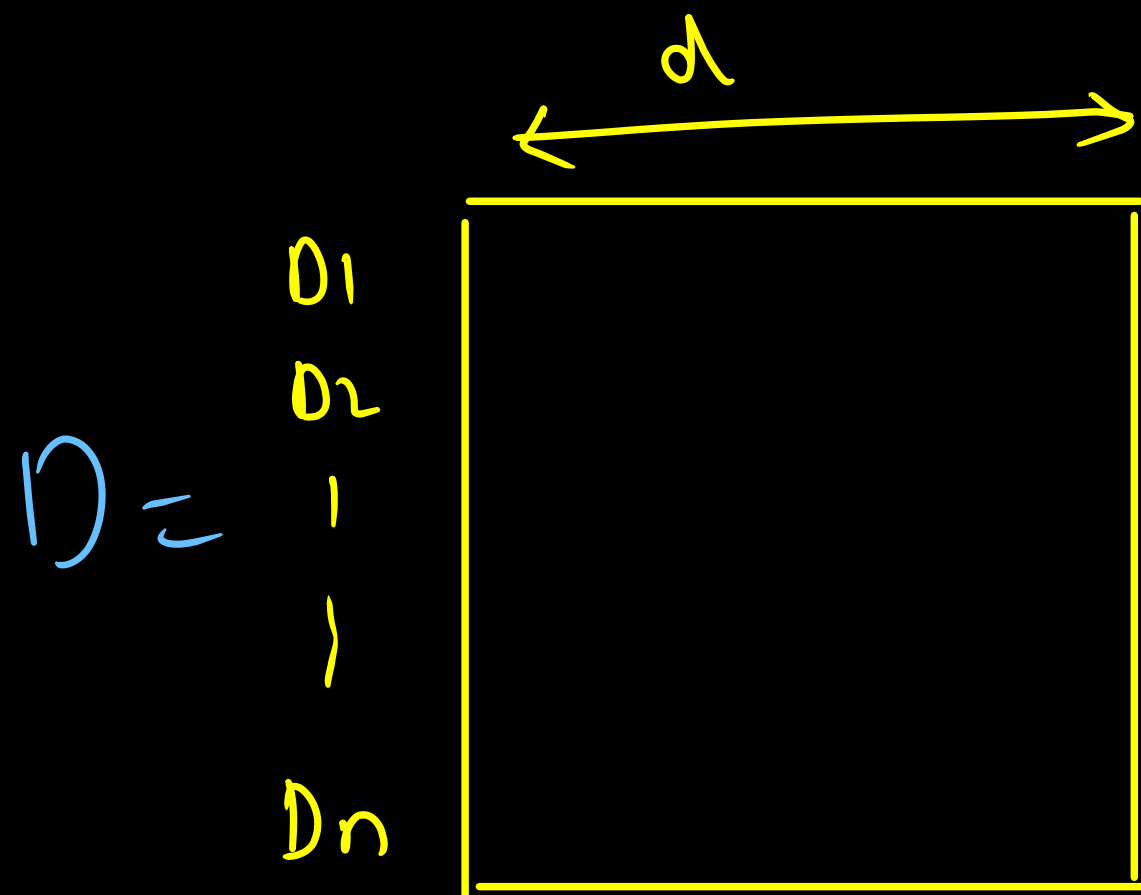
↓
freq of each word
in document

Doc-Word
Matrix ←

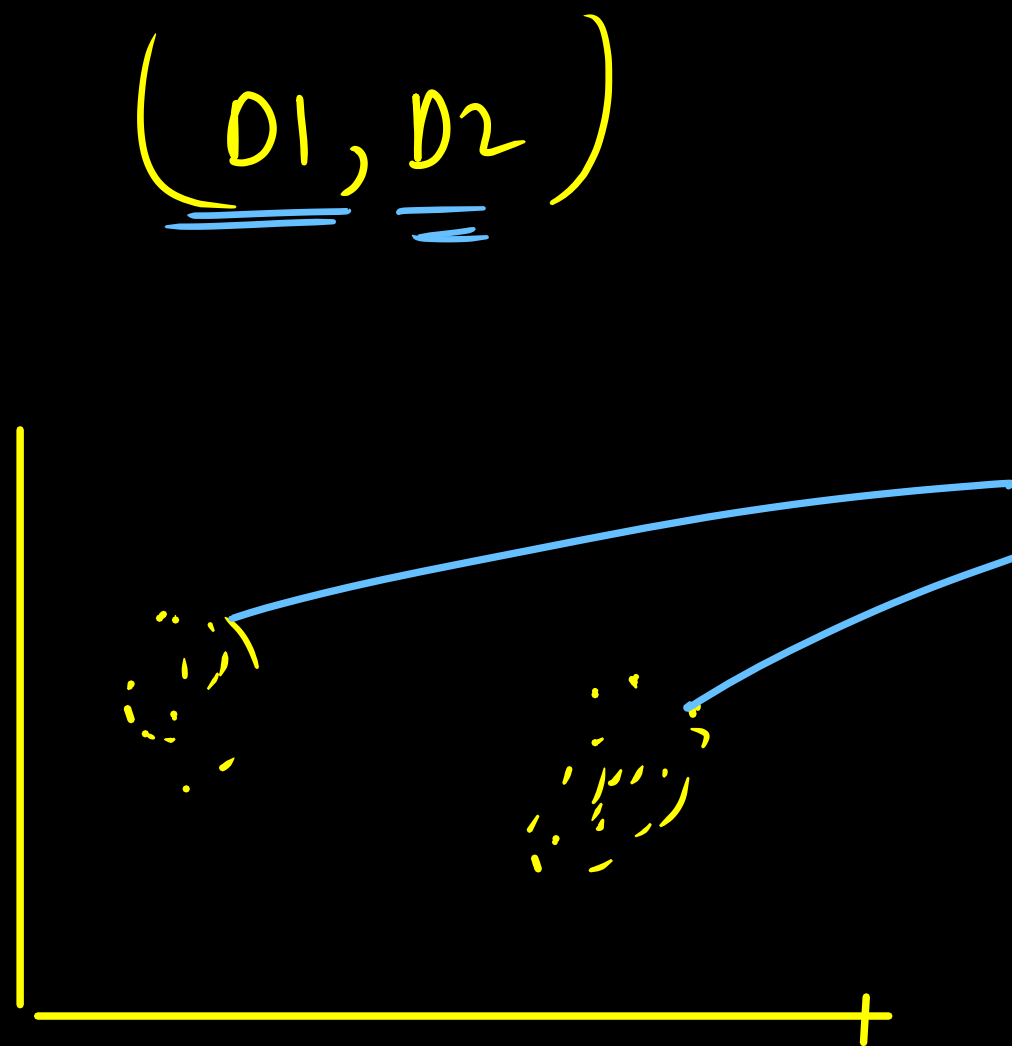
$$A_{n \times m} = B_{n \times d} C_{d \times m}^T$$

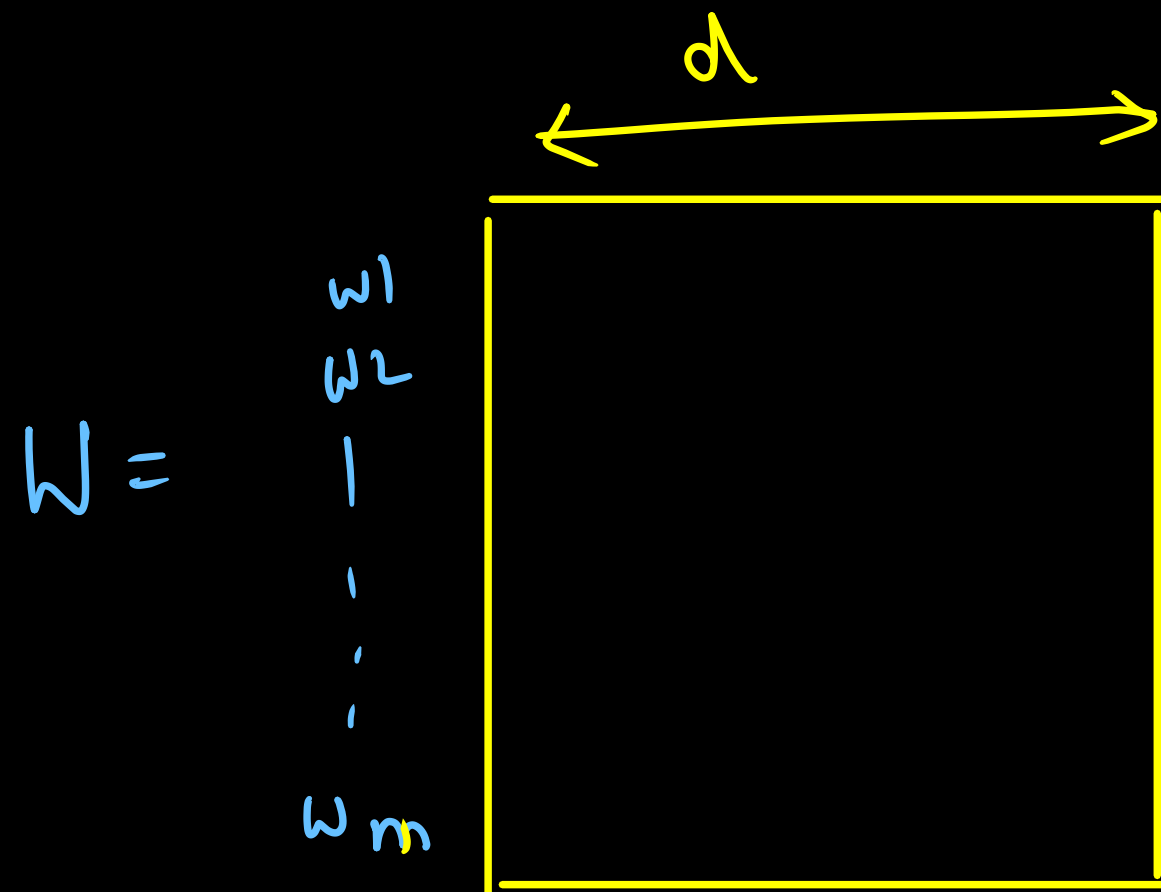
B = Doc. matrix in d-dimension

C = Word. matrix in d-dimension



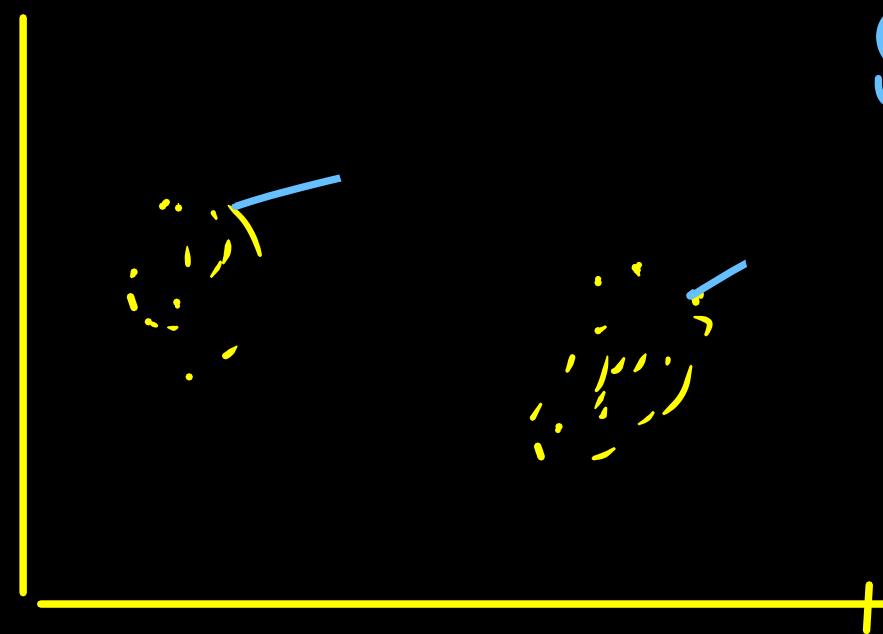
$D D^T$





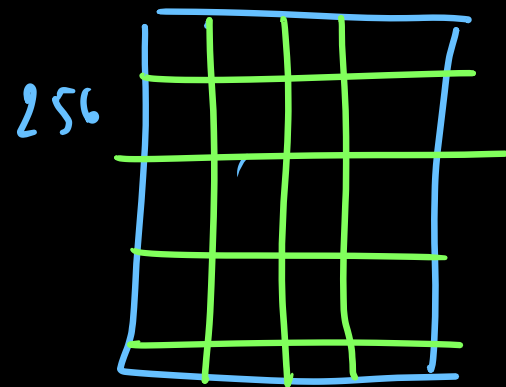
$$D D^T$$

(w_1, w_2)



Similar words
will have
higher cosine
similarity.

PCA → Eigen Face

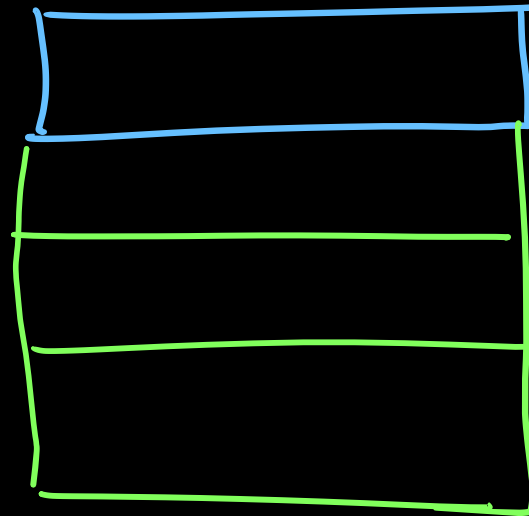


m

✓ ✓
 $X (m, 65K)$

$$256 \times 256 = 2^{16} = 65536$$

$$= 65K$$



✓
 $m \times 65K$

$$X^T X \rightarrow (65K, 65K)$$

top-K

✓ ✓
 $\tilde{X}^2 = X W$
 $m \times K \quad m \times 65K \quad 65K \times K$
✓ ✓
✓ $m \times 100$ ✓

~ 90% Variance

$(256, 256)$



flatten